IEOR 221

Homework 4

Module 5 Options – Basics (September 29, 2025)

1) A 1-month European put option on a non-dividend-paying stock is currently selling for \$2.50. The stock price is \$47, the strike price is \$50, and the risk-free interest rate is 6% per annum. What opportunities are there for an arbitrageur? (Hull 11.17)

2) The price of a European call that expires in 6 months and has a strike price of \$30 is \$2. The underlying stock price is \$29, and a dividend of \$0.50 is expected in 2 months and again in 5 months. Risk-free interest rates (all maturities) are 10%. What is the price of a European put option that expires in 6 months and has a strike price of \$30? (Hull 11.19)

3) Suppose that c_1, c_2, and c_3 are the prices of European call options with strike prices K_1, K_2, and K_3, respectively, where $K_3 > K_2 > K_1$ and $K_3 - K_2 = K_2 - K_1$. All options have the same maturity. Show that

$$c_2 \leq \frac{c_1 + c_3}{2}$$

(Hint: Consider a portfolio that is long one option with strike price K_1, long one option with strike price K_3, and short two options with strike price K_2.) (Hull 11.31)

4) A trader creates a bear spread by selling a 6-month put option with a \$25 strike price for \$2.15 and buying a 6-month put option with a \$29 strike price for \$4.75. What is the initial investment? What is the total payoff (excluding the initial investment) when the stock price in 6 months is (a) \$23, (b) \$28, and (c) \$33. (Hull 12.22)

5) A diagonal spread is created by buying a call with strike price K_2 and exercise date T_2 and selling a call with strike price K_1 and exercise date T_1, where $T_2 > T_1$. Draw a diagram showing the profit from the spread at time T_1 when (a) $K_2 > K_1$ and (b) $K_2 < K_1$. (Hull 12.25)